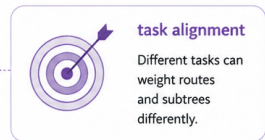
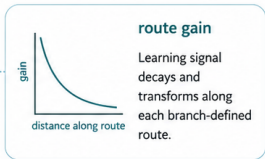
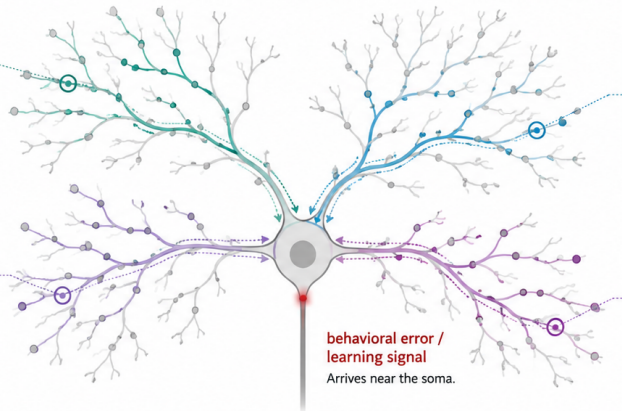
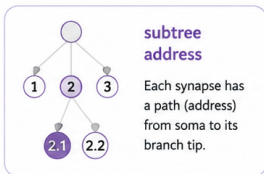
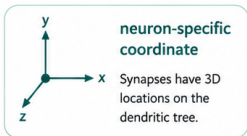
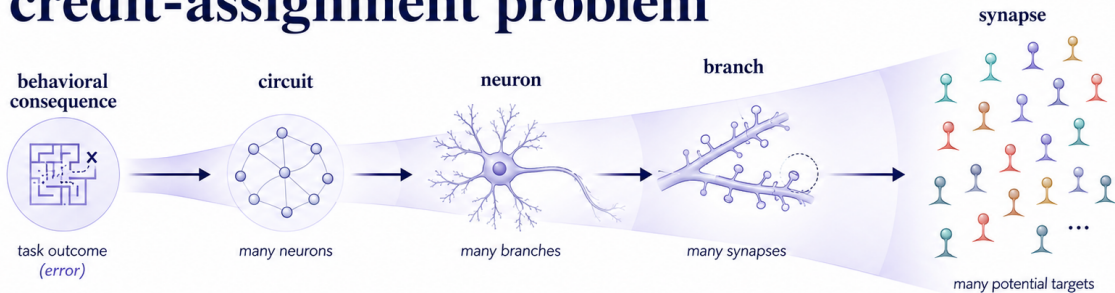


When dendritic structure helps local credit assignment

Coordinates, subtree addresses, route gain, and task alignment



Learning in the brain is a credit-assignment problem

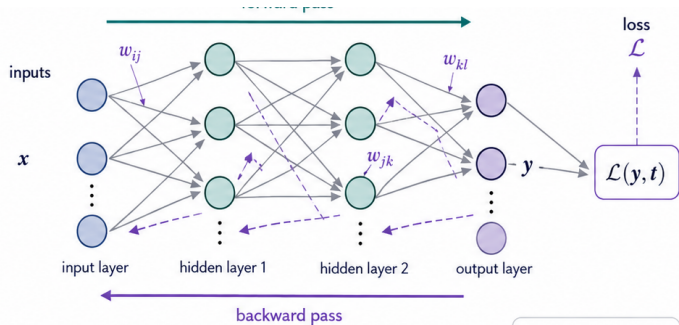


One behavioral error must be assigned to many local causes.



Sign, magnitude, and destination of credit are all needed.

Backpropagation solves parameter credit in artificial networks



$$\frac{\partial \mathcal{L}}{\partial w_{ij}} = \left(\frac{\partial \mathcal{L}}{\partial y_j} \right) \left(\frac{\partial y_j}{\partial w_{ij}} \right)$$



exact signed update



every parameter gets a destination

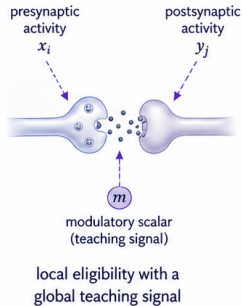


global coordination across layers

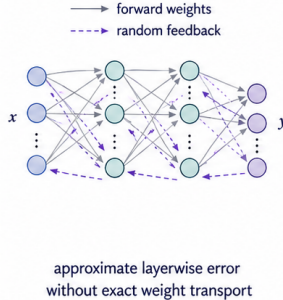
Backpropagation is the optimization reference because it tells every weight how it changed the loss.

Known biological learning theories solve different parts of credit assignment

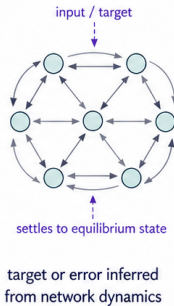
three-factor plasticity



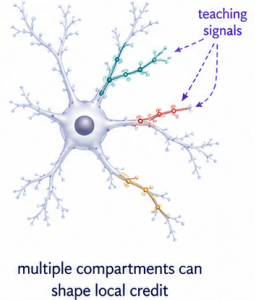
feedback alignment



predictive / equilibrium methods

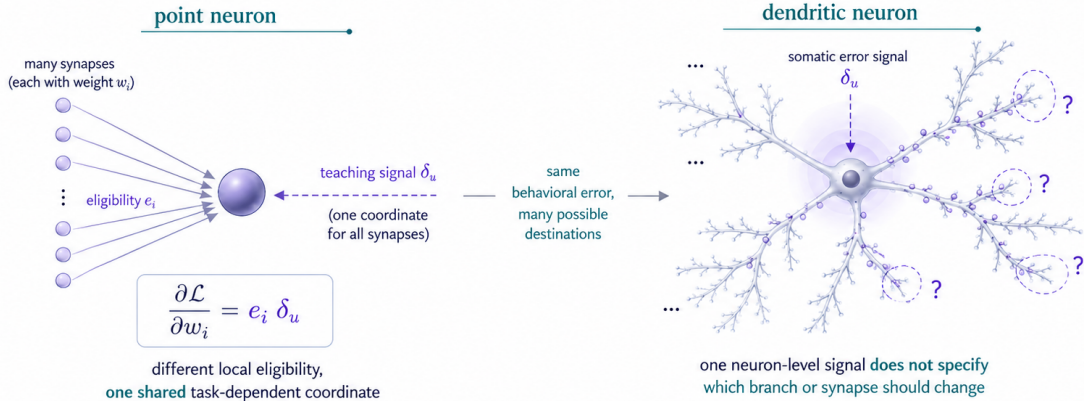


dendritic / compartment models



Most approaches provide a neuron-level teaching coordinate or a few compartment signals; the fine within-tree destination problem remains open.

The unresolved problem begins inside the neuron



Within-neuron routing is the missing spatial step beyond neuron-level credit.

A dendritic tree adds local state, subtree addresses, and route gain

1 local state

Voltage/conductance dependence at compartments or branches.

$V, g \rightarrow$ nonlinear, state-dependent integration.



2 subtree address

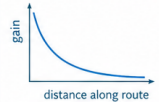
Soma-to-branch path hierarchy or ancestry route.

① → ② → ③ → ...

Encodes location and topology within the dendritic tree.

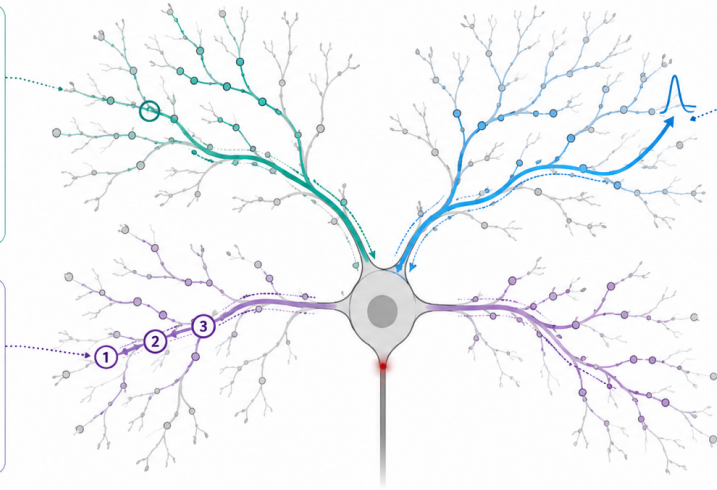
3 route gain

Attenuation or transformation of the signal along a branch-defined path.



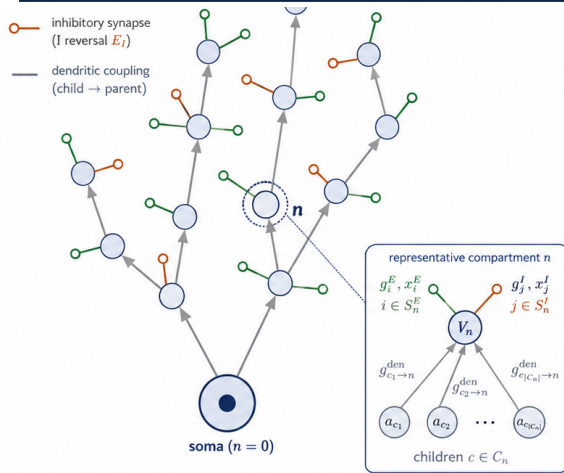
4 task alignment

Task demands determine whether local state, addresses, and route gain improve learning and performance.



Dendrites add spatial structure and state dependence beyond the point-neuron model.

A dendritic neuron is a conductance tree w



$$\mathcal{O}_n, \mathcal{O}_n, \mathcal{O}_n = \mathcal{O}_n \uplus \mathcal{O}_n$$

$$G_n^E = \sum_{i \in S_n^E} g_i^E x_i^E$$

$$G_n^I = \sum_{j \in S_n^I} g_j^I x_j^I$$

$$G_n^{\text{child}} = \sum_{c \in C_n} g_{c \rightarrow n}^{\text{den}}$$

$$J_n^{\text{child}} = \sum_{c \in C_n} g_{c \rightarrow n}^{\text{den}} a_c$$

2) Current balance and steady-state solution

$$C_n \frac{dV_n}{dt} = g_n^L (E_L - V_n) + G_n^E (E_E - V_n) + G_n^I (E_I - V_n) + \sum_{c \in C_n} g_{c \rightarrow n}^{\text{den}} (a_c - V_n)$$

$$g_n^{\text{tot}} = g_n^L + G_n^E + G_n^I + G_n^{\text{child}}$$

$$R_n^{\text{tot}} = \frac{1}{g_n^{\text{tot}}}$$

$$V_n = R_n^{\text{tot}} (g_n^L E_L + G_n^E E_E + G_n^I E_I + J_n^{\text{child}})$$

3) Normalized leaf approximation

$$g_n^L = 1, \quad E_L = E_I = 0, \quad E_E = 1$$

$$V_{\text{leaf}} = \frac{G^E}{1 + G^E + G^I}$$

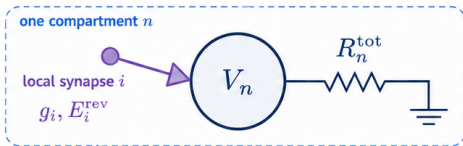
4) Interpretation

Excitatory and inhibitory synapses both add positive conductance to the denominator; their different reversal potentials determine the sign of current.

Exact local eligibility with explicit E/I synapses

Exact gradient factorization at synapse i on compartment n

$$\begin{aligned}\frac{\partial L}{\partial g_i} &= e_i^{\text{den}} \delta_{n,u}^V \\ e_i^{\text{den}} &= x_i R_n^{\text{tot}} (E_i^{\text{rev}} - V_n) \\ \delta_{n,u}^V &= \frac{\partial L}{\partial V_n}\end{aligned}$$



Local eligibility factors e_i^{den}

presynaptic activity

$$x_i^X$$

transmitter release
or spike-dependent
term

local input resistance

$$R_n^{\text{tot}}$$

total input resistance
seen at compartment n

driving force

$$(E_X - V_n)$$

difference between
reversal potential of
class X and local voltage

Sensitivity of V_n to a synaptic conductance of class X

$$\frac{\partial V_n}{\partial g_i^X} = x_i^X R_n^{\text{tot}} (E_X - V_n), \quad X \in \{E, I\}$$

Explicit excitatory and inhibitory gradient cases

$$\frac{\partial L}{\partial g_i^E} = x_i^E R_n^{\text{tot}} (E_E - V_n) \delta_{n,u}^V, \quad i \in \mathcal{S}_n^E$$

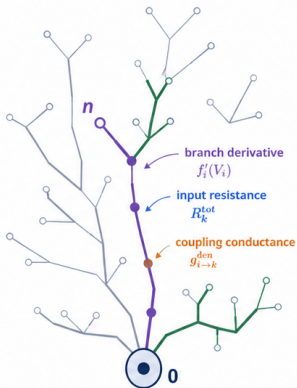
$$\frac{\partial L}{\partial g_j^I} = x_j^I R_n^{\text{tot}} (E_I - V_n) \delta_{n,u}^V, \quad j \in \mathcal{S}_n^I$$

Interpretation

- E and I have opposite local voltage sensitivities when $E_I < V_n < E_E$.
- The sign of the final loss gradient also depends on the compartment learning signal $\delta_{n,u}^V$.
- Near E_I , an inhibitory synapse can weakly drive its own current yet still change other gradients through conductance loading.

From tree transport to the general adjoint

Directed tree: single path transport of returned signal



Recursive transport along the unique path from n to 0

$$\delta_{0,u}^V = f'_0(V_0) \delta_u$$

$$\delta_{n,u}^V = \delta_{p(n),u}^V f'_n(V_n) R_{p(n)}^{\text{tot}} g_{n \rightarrow p(n)}^{\text{den}}$$

$$\delta_{n,u}^V = \delta_{0,u}^V \tilde{\alpha}_n$$

$$\tilde{\alpha}_n = \prod_{(i \rightarrow k) \in \text{path}(n \rightarrow 0)} f'_i(V_i) R_k^{\text{tot}} g_{i \rightarrow k}^{\text{den}}$$

$f'_i(V_i)$: branch derivative

R_k^{tot} : input resistance

$g_{i \rightarrow k}^{\text{den}}$: coupling conductance

$p(n)$

parent of node n on the path to the root

$\text{path}(n \rightarrow 0)$

ordered set of directed edges from n to root 0

$(n = i_0 \rightarrow i_1 \rightarrow \dots \rightarrow i_m = 0)$

General compartmental system: steady-state adjoint

$$F(V, g, x) = 0$$

steady-state residual
(nonlinear in general)

$$J_V = \frac{\partial F}{\partial V}$$

Jacobian w.r.t. V

$$J_V^T q = \nabla_V L$$

adjoint linear system

$$\frac{\partial L}{\partial g_i} = -q^T \frac{\partial F}{\partial g_i}$$

gradient of scalar objective L
(adjoint sensitivity formula)

For the standard residual convention

$$F = GV - b$$

nodal conductance matrix G , source b

$$\frac{\partial L}{\partial g_i} = x_i (E_i^{\text{rev}} - V_n) q_n$$

local sensitivity (edge i from n to m)

Relationship to directed-tree transport

$$q_n = R_n^{\text{tot}} \delta_{n,u}^V$$

adjoint equals input resistance
times returned signal

In a directed tree, error transport is a single path product; in a general compartmental system, it is carried by the full adjoint / Green's function.

Restricted routed local learning rule

Exact and approximate local rules

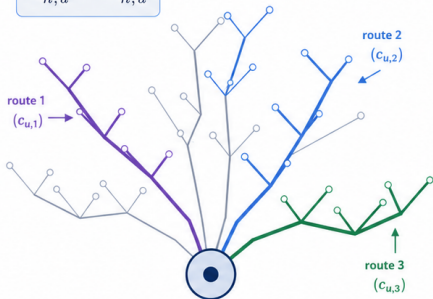
Exact approximate-gradient form:

$$\hat{\nabla}_{g_i} L = x_i R_n^{tot} (E_i^{rev} - V_n) \delta_{n,u}^V$$

Exact

Local approximate-gradient form:

$$\delta_{n,u}^V = \delta_{n,u}$$



Routed teaching field (approximate local feedback field)

$$\delta_{n,u}^V = \sum_{k=1}^K A_{u,nk} c_{u,k}$$

$A_u \in \mathbb{R}^{N_u \times K}$ fixed route/address matrix

$c_u = (c_{u,1}, \dots, c_{u,K})^T$ communicated route coefficients

K within-neuron feedback bandwidth

N_u number of compartments in neuron u

	route 1	route 2	...	route K
compartment 1	1	0	...	0
compartment 2	1	0	...	0
compartment 3	0	1	...	0
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
compartment N_u	0	0	...	1

Route coefficients c_u (length K)

$$\begin{bmatrix} c_{u,1} \\ c_{u,2} \\ \vdots \\ c_{u,K} \end{bmatrix}$$

each value is assigned to the compartments of its route

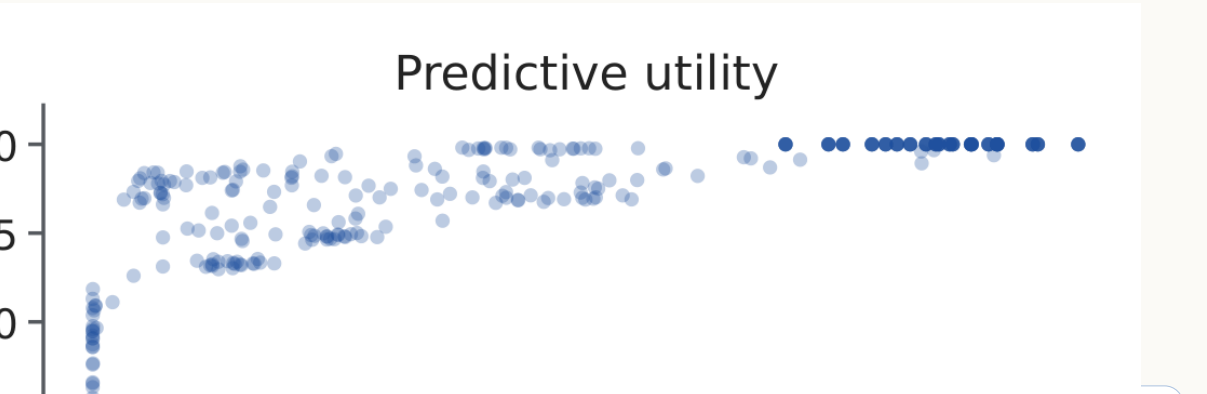
Bandwidth and routing choices: from scalar to exact compartment field

1) Layer-wide scalar	2) Neuron-specific feedback	3) Addressed routed field	4) Exact compartment field
one value shared by all neurons	one value per neuron, repeated within that neuron	several values per neuron, routed to selected subtrees	one independent value per compartment
bandwidth $K = 1$ shared across layer	bandwidth $K = 1$ per neuron	bandwidth $K > 1$ per neuron (routed)	bandwidth $K = N_u$ (exact)

Interpretation

The dendritic eligibility remains exact and local; *learning quality depends on how well the available routed field captures the exact credit field.*

A stochastic credit operator predicts wh



The theory predicts both benefits and nulls from the geometry of the available credit

Neuron identity dominates standard tasks

On standard tasks, the major bottleneck for local credit assignment is selecting the correct neuron, not resolving the exact path.

Tasks and setup

Two standard tasks

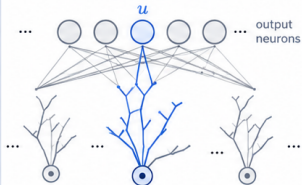
MNIST



Flattened CIFAR-10



Network schematic (final layer)



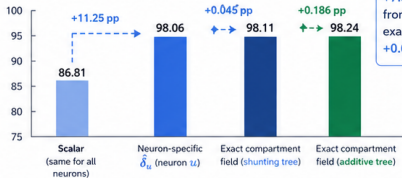
u = target neuron

Training signal must be routed back into neuron u .

Compare scalar, neuron-specific, and exact compartment feedback under matched forward models.

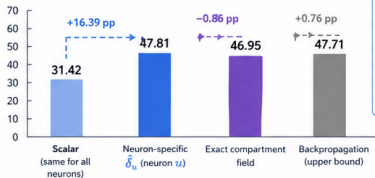
Performance comparison (higher is better)

MNIST (test accuracy, %)



+11.25 pp in shunting tree; +7.81 pp in additive tree from neuron identity; exact field adds only +0.045 / +0.186 pp.

Flattened CIFAR-10 (test accuracy, %)



Neuron-specific feedback adds +16.39 pp over scalar; the exact compartment field is 0.86 pp below neuron-specific and is equivalent to backpropagation within a predefined 1-point margin.

Takeaways

- Correct **neuron identity** is the main missing coordinate.
- Exact path resolution **matters little** on standard tasks.
- This motivates tasks with **within-neuron credit conflict**.

Notation

u = target neuron (the correct output neuron for the training example).

$\hat{\delta}_u$ = neuron-specific learning signal for neuron u , identical across all compartments of u .

ed forward architectures; the topology-matched raw-additive control removes the conductance denominator while preserving the

Credit conflict creates a demand for bra

compatible, $\chi = 0$

$x_b : y$

$b = 1$

$x_c : y$

$b = 2$

$x_b : y$

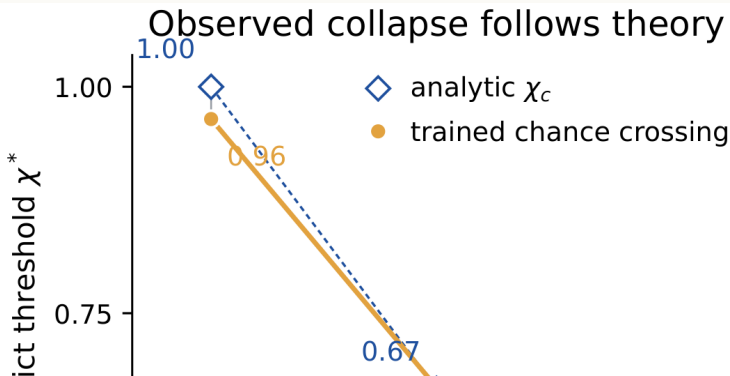
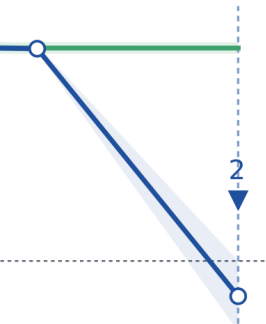
$b = 3$

downstream gate c

need for an internal branch address when one neuron must assign opposing credit

Shared credit fails at the predicted conf

d boundary



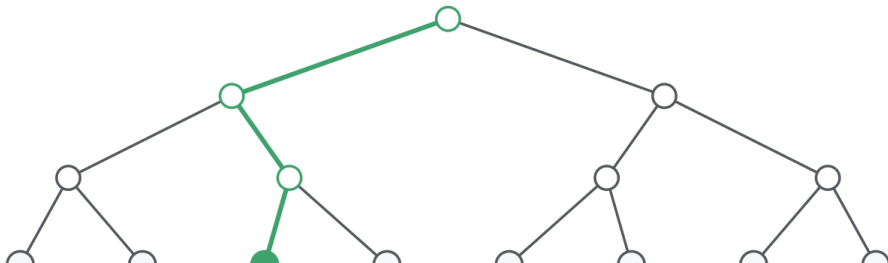
credit becomes useful precisely when a shared neuronal signal cancels across com

The right subtree address matters; fine

A

Eight-context hierarchical task

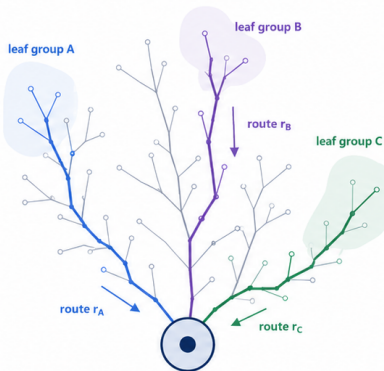
context $c = 3$ selects stream 3; selected stream carries $+y$; all streams active



is essential, while the extra benefit of this particular tree topology remains conditional

Real arbors provide sparse candidate routes

1) Reconstructed arbor with ancestry-defined routes



Measured anatomy, modeled credit fields.

2) Route capture: definition and results (8 channels)

Capture of a route dictionary A

$$C_A(q) = \frac{\|P_A q\|^2}{\|q\|^2}$$

Capture is the fraction of modeled credit-field energy contained in the span of the route dictionary A .

Results at 8 channels

Subtree routes retain of dense rank-matched capture	85%
Using about of dense route-matrix connections	6.9%
Dense capture per connection (subtree vs. dense)	14.2×
Advantage over shuffled dictionary at matched density	≈ 2.7×

Dense: rank-matched optimal dense dictionary.
Shuffled: random dictionary with the same density.

3) Interpretation

- Reconstructed arbors provide sparse candidate route dictionaries.

Ancestry-defined subtree routes capture most of the modeled credit-field energy while using very few potential routing connections.

- These are modeled fields on measured anatomy, not observed task gradients.

Capture is computed from operator-induced credit fields on the reconstructed morphology. No task gradients were used.

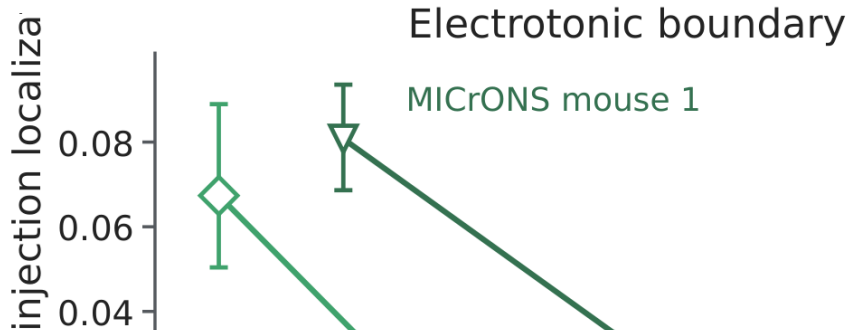
- Ordering replicates in a disjoint 47-cell cohort.

The same ordering (subtree > dense at matched density) 47 cells

Same ordering in a descriptive 10-cell second-mouse cohort

Across a disjoint 47-cell cohort, real arbors yield dictionaries that are far sparser yet much more capture-efficient than random alternatives.

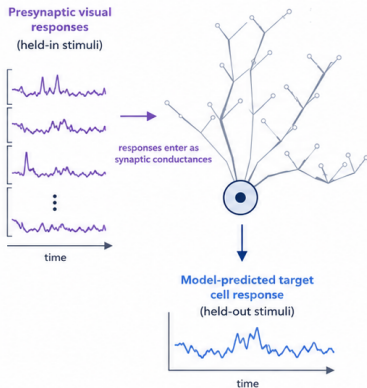
Focal shunting changes descendant credit



ulate an already-addressed credit route, but ordinary passive parameters do not gu

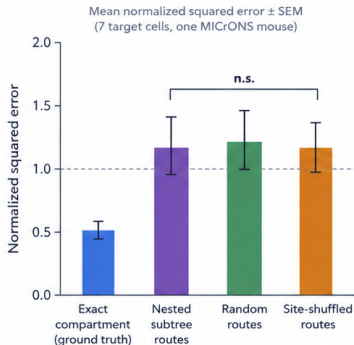
Measured visual responses show no morphology-specific alignment

MICrONS response-prediction setup



We use presynaptic visual responses (held-in) to drive conductances on the reconstructed arbor and predict the target cell's response to held-out stimuli.

Held-out prediction error (lower is better)



Across 7 target cells from one MICrONS mouse, nested subtree routes do not beat random or site-shuffled routes for held-out prediction.

Interpretation

No significant morphology-specific advantage in held-out prediction.

Also no advantage in response-derived field capture or within-arbor structure-function similarity.

This is the strongest boundary on the biological claim.

from one MICrONS mouse: topology-minus-matched-control effects cross zero. Capacity does not establish endog

Imposed alignment makes the same and

$$\phi(a) = \sqrt{a} u_{\parallel} + \sqrt{1-a} u_{\perp}, \quad C_A[\phi(a)] = a$$

Imposed alignment

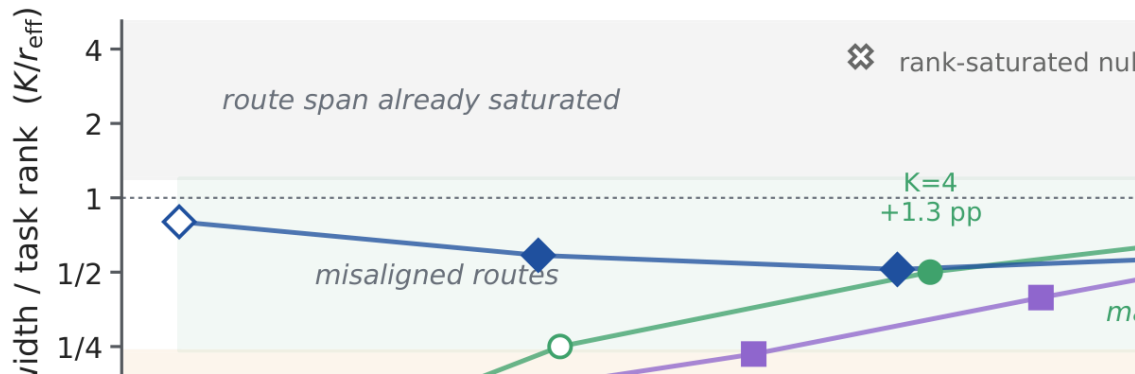
$a = 0$

$a = 0.4$

$a = 1$

Technology becomes useful when the task-credit field lies in the span of its available ro

Alignment and bandwidth organize the



is a conditional substrate for routing local credit—not a general replacement for